

PRITHWIRAJ BHATTACHARJEE

Assistant Professor of Computer Science and Engineering

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Research Interests

Large language model (LLM) reasoning and reliability; knowledge-grounded retrieval (Graph RAG); zero-shot learning; low-resource natural language processing; prompt adaptation and optimization; and explainable AI.

My research asks why models behave the way they do, not only whether they perform well. My work spans zero-shot text classification, Bangla language technologies, abstractive summarization with supervised fine-tuning and reinforcement learning, and, more recently, knowledge graphs and retrieval-augmented generation for grounding LLM outputs. I treat interpretability as a design constraint rather than something added after the fact. My broader work also includes computer vision, multimodal systems, and explainable modeling for time-series data.

Education

M. Sc. in Computer Science and Engineering

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

GPA:3.85/4.00; ranked 1st of 7.

Oct. 2020 – Dec. 2021

B. Sc. (Engg.) in Computer Science and Engineering

Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh

GPA:3.86/4.00; ranked 3rd of 62.

Dec. 2015 – Dec. 2019

Academic Appointments

Assistant Professor, Department of Computer Science and Engineering

Leading University, Sylhet, Bangladesh

Nov. 2025 – Present

- Principal investigator for 10+ concurrent research projects in NLP, multimodal AI, clinical screening, and explainable AI.
- Teach Advanced Machine Learning, Artificial Intelligence, Natural Language Processing, and Numerical Methods.
- Convener, Departmental Routine Committee.

Lecturer, Department of Computer Science and Engineering

Leading University, Sylhet, Bangladesh

Feb. 2021 – Oct. 2025

- Taught Data Structures, Computer Algorithms, Object-Oriented Programming, Python, and Discrete Mathematics.

Adjunct Lecturer, Department of Computer Science and Engineering

Sylhet International University, Sylhet, Bangladesh

Jul. 2020 – Dec. 2020

ACM Programming Instructor and Head Coach

Leading University, Sylhet, Bangladesh

May 2022 – Oct. 2024

- Mentored 100+ students in advanced algorithmic problem solving.
- Head coach for teams at the ACM ICPC Dhaka Regional (2021, 2023), JU NCPC (2023), IUT IUPC (2024), and BUET IUPC (2023).

Research Experience

LLMs, NLP, and Language Technologies

- Studying calibration and label-free threshold selection for zero-shot topic and multi-label text classification.

- Developing sequential SFT and reinforcement learning (SCST, PPO) pipelines for low-resource Bangla abstractive summarization, and weakly supervised deep clustering for low-resource emotion detection.
- Built neural abstractive summarization models and datasets for Bangla, including attention-based and pointer-generator models.
- Exploring knowledge graphs and Graph RAG for knowledge-grounded, reliable LLM reasoning.

Computer Vision, Multimodal AI, and Explainable ML

- Supervise projects on synthetic-media detection, plant-disease detection and localization, flower recognition, and road-hazard detection.
- Build explainable (SHAP-based) models for time-series and tabular problems in environmental risk, finance, agriculture, and clinical screening.

Publications and Manuscripts

An asterisk () denotes the corresponding author, as indicated in the original publication records. Within each group, papers are ordered by my author position.*

LLMs, NLP, and Language Technologies

Published

- [1] **Bengali Abstractive News Summarization (BANS): A Neural Attention Approach**
Prithwiraj Bhattacharjee*, Avi Mallick, Md. Saiful Islam, and Mariam Jannat.
Proceedings of the International Conference on Trends in Computational and Cognitive Engineering, Springer AISC vol. 1309, 2021.
[Paper link](#)
- [2] **End-to-End Bangla Speech Synthesis**
Prithwiraj Bhattacharjee*, Rajan Saha Raju, A. Ahmad, and Md. Shahidur Rahman.
International Conference on Science and Contemporary Technologies (ICSCT), IEEE, 2021.
[Paper link](#)
- [3] **A Bangla Text-to-Speech System Using Deep Neural Networks**
Rajan Saha Raju, Prithwiraj Bhattacharjee, A. Ahmad, and Md. Shahidur Rahman.
International Conference on Bangla Speech and Language Processing (ICBSLP), IEEE, 2019.
[Paper link](#)
- [4] **MultiBanAbs: A Comprehensive Multi-Domain Bangla Abstractive Text Summarization Dataset**
Md. Tanjim Ferdous, Naeem Ahsan Chowdhury, and Prithwiraj Bhattacharjee*.
Trends in Computational and Cognitive Engineering, Springer LNNS, 2026.
[Paper link](#)
- [5] **Pointer over Attention: An Improved Bangla Text Summarization Approach Using Hybrid Pointer Generator Network**
Nobel Dhar, Gaurob Saha, Prithwiraj Bhattacharjee, Avi Mallick, and Md. Saiful Islam.
24th International Conference on Computer and Information Technology (ICCIT), IEEE, 2021.
[Paper link](#)
- [6] **SentiTube: YouTube Comment-Based Bangla Sentiment Analysis Dataset**
Asfakur Rahman, Amer Mahbub, Biswa Nikhil Paul, Prithwiraj Bhattacharjee*, and Md. Akhtaruzzaman.
7th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI), 2025.
[Paper link](#)

Under review and submitted

- [7] **Keyword Augmentation as Calibration in Zero-Shot Topic Classification.**
Prithwiraj Bhattacharjee.
Under review. ECTI Transactions on Computer and Information Technology (ECTI-CIT).
- [8] **A Label-Free Method for Choosing Per-Topic Decision Thresholds in Zero-Shot Multi-Label Text Classification.**
Prithwiraj Bhattacharjee.
Under review. Journal of Research and Development.
- [9] **Bridging the SFT–RL Gap: Sequential Policy Optimization for Low-Resource Bangla Text Summarization.**
Poroma Chowdhury, Ibsar Ahmed, and **Prithwiraj Bhattacharjee***.
First revision. ACM Transactions on Asian and Low-Resource Language Information Processing (Q2).
- [10] **Beyond Zero-Shot: Efficient Low-Resource Emotion Detection via Weakly Supervised Deep Clustering.**
Abdul Amin, Nuzahat, and **Prithwiraj Bhattacharjee.**
Submitted. International Conference on Computer and Information Technology (ICCIT), 2026.

Computer Vision and Multimodality

Published

- [11] **BloomNet: Exploring Single vs. Multiple Object Annotation for Flower Recognition Using YOLO Variants**
Safwat Nusrat and **Prithwiraj Bhattacharjee***.
Trends in Computational and Cognitive Engineering, Springer LNNS, 2026.
[Paper link](#)
- [12] **Toward Generalized Detection of Synthetic Media: Limitations, Challenges, and the Path to Multimodal Solutions**
Redwan Hussain, Mijanur Rahman, and **Prithwiraj Bhattacharjee***.
Trends in Computational and Cognitive Engineering, Springer LNNS, 2026.
[Paper link](#)
- [13] **RoadGuard: A Real-Time Mobile-Based Road-Hazard Detection, Warning, and Hazard-Aware Route Recommendation System for Road Users and Vehicles**
A. S. Mashrafi, M. A. S. Rabbi, A. T. Rajib, and **Prithwiraj Bhattacharjee.**
Systems and Soft Computing, Elsevier (Q2).
[Paper link](#)
- [14] **LightTeaNet: A Weakly Supervised Lightweight CNN for Multi-Label Tea Leaf Disease Detection and Localization**
Naif Haider Chowdhury, Md. Abdur Rahim, S. F. Hasan, Murad Hasan, and **Prithwiraj Bhattacharjee.**
Neural Computing and Applications, Springer, 2026 (Q1).
[Paper link](#)

Under review and submitted

- [15] **GenDet-M: A Multimodal Ensemble Architecture for Detecting High-Fidelity Synthetic Video.**
Redwan Hussain, Mijanur Rahman, and **Prithwiraj Bhattacharjee***.
Under review. Neural Computing and Applications, Springer, 2026 (Q1).

AI and Time-Series Explainability

Published

- [16] **AgriCast: Weather-Based Forecasting of Staple Crop Prices with Machine Learning and Deep Learning**
Nirab Debnath, Parishmita Sinha, and **Prithwiraj Bhattacharjee***.
28th International Conference on Computer and Information Technology (ICCIT), IEEE, 2025.
[Paper link](#)

Under review and submitted

- [17] **AeroHealth: An Explainable AI Framework for Assessing PM_{2.5} Health Risk.**
Mohuya and **Prithwiraj Bhattacharjee**.
Under review. Environmental Monitoring and Assessment (Q2).
- [18] **Hydroclimatic Drivers of River Degradation: An Explainable Ensemble Learning Framework for Water Quality in Dhaka.**
Parishmita Sinha, Nirab Debnath, and **Prithwiraj Bhattacharjee***.
First revision. Water, Air, & Soil Pollution (Q2).
- [19] **A Bi-GRU Tri-modal Framework for Ethereum Price Prediction for Multi-day Horizon.**
Dip Barman, Nazmul Tahsin, and **Prithwiraj Bhattacharjee**.
Under review. SN Computer Science (Q2).
- [20] **Socioeconomic Equity of Tea Garden Workers: A Composite Financial Inclusion Index via Explainable AI.**
Nirob, Choity, and **Prithwiraj Bhattacharjee**.
Submitted. International Conference on Computer and Information Technology (ICCIT), 2026.

Ongoing Research Projects

In progress; no manuscript submitted yet.

- **Struct2Test:** A graph neural network approach for automated UI test-script generation from Android layouts.
- **Emoji Recommendation for Educational Notifications:** HCI-driven dataset construction and user-centric recommendation modeling.
- **Data-Driven Antenatal Care:** Explainable AI for complication detection and treatment prediction on a proprietary 500-patient clinical dataset.
- **CrossASD:** A fair, minimal-feature machine-learning framework for autism spectrum disorder screening.

Teaching and Mentoring

Selected Courses Taught

- Advanced Machine Learning
- Artificial Intelligence
- Natural Language Processing
- Data Structures and Algorithms
- Object-Oriented Programming and Python
- Discrete Mathematics and Numerical Methods

Student Research Supervision

Principal investigator for 10+ concurrent student research projects across NLP, multimodal AI, computer vision, clinical screening, and explainable machine learning.

Competitive Programming Coaching

Mentored 100+ students in algorithmic problem solving and prepared teams for national and regional programming contests.

Academic Service and Leadership

- Convener, Departmental Routine Committee, Leading University.
- Problem Setter and Judge, Debug Wars, IEEE Computer Society LU Chapter, 2026.
- Problem Setter and Judge, Intra-LU Programming Contest, LU Computer Club, 2024.
- Former General Secretary, IEEE Computer Society SUST Student Branch Chapter.

Technical Skills

Machine Learning and AI: PyTorch, TensorFlow, Hugging Face Transformers, LLM fine-tuning (LoRA/Adapters, PPO, RLHF), RAG, Graph RAG, knowledge graphs, SHAP analysis, and statistical testing.

Computer Vision: YOLO, CNNs, CAM, object detection, image classification, localization, multimodal fusion, and TensorFlow Lite.

Programming: Python, C/C++, Java, JavaScript, PHP, and C#.

Tools and Platforms: Git, Docker, Linux, LaTeX, Django, Laravel, ReactJS, ExpressJS, and Unity.

Databases: MySQL, MongoDB, and Firebase.

Honors and Additional Information

- Talent Pool Board Scholarship, Bangladesh Board of Education, 2013; ranked 2nd in district.
- Government Board Scholarships, Bangladesh Board of Education, 2007, 2010, and 2015.
- Solved more than 500 programming problems across online judges.

Languages: English (IELTS Academic, overall band 7.0: Speaking 7.5, Listening 7.5, Reading 7.0, Writing 6.0); Bengali (native).

References

Dr. Mohammad Shahidur Rahman, Professor, Department of CSE, Shahjalal University of Science and Technology, Sylhet. rahmanms@sust.edu

Dr. Md Sadek Ferdous, Professor, Department of CSE, Brac University, Dhaka. sadek.ferdous@bracu.ac.bd